

Program: Electronics and Computer Engineering	
Course Code: 3509	Course Title: Basics of C Programming Lab
Semester: 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To make use of programming fundamentals using C language.
- To build a strong foundation to learn other programming languages.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic principles and theorems of Engineering Mathematics	1002, 2002	Mathematics I &II	1 & 2
Basic functions and features of Computer, Operating system and Internet applications, basic programming skills in Python.	1008	Introduction to IT systems Lab	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Make use of various data types, I/O statements and basic operators in C programming	11	Applying
CO2	Apply decision making and loop statements in C programming.	13	Applying
CO3	Develop C programs based on arrays.	9	Applying
CO4	Develop C programs using pointers, string handling functions and user-defined functions.	9	Applying
	Series Test	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3		3	3			
CO3	3	3	3	3			
CO4	3	3	3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Make use of various data types, I/O statements and basic operators in C programming		
M1.01	Demonstrate output and input functions.	5	Understanding
M1.02	Demonstrate different fundamental data types in C.	3	Understanding
M1.03	Solve simple problem statements using operators.	3	Applying
	Suggested experiments 1. Write a C program to perform addition, subtraction, division and multiplication of two numbers. 2. Write a program to find the roots of quadratic equation. Write a Program to read height, width and depth of a cube, then calculate and display the volume of the Cube.		
CO2	Apply decision making and loop statements in C programming		
M2.01	Solve problem statements in C using flow control statements - if, if-else, nested if, switch.	8	Applying
M2.02	Solve problem statements using loop statements - while, do ... while and for loop.	5	Applying
	Series Test 1	1.5	

	Suggested Experiments 1. Write a program to find the largest and smallest among three entered numbers and also display whether the identified largest/smallest number is even or odd. 2. Write a program to find whether a character is consonant or vowel using switch statement 3. Write a program to generate Fibonacci series.		
CO3	Construct C programs based on arrays.		
M3.01	Make use of one dimensional array operations - largest, smallest and compute sum & average of array elements	3	Applying
M3.02	Write programs to perform two dimensional array operations - transpose of matrix, matrix addition, matrix subtraction	6	Applying
	Suggested Experiments 1. Write a Program find the sum of elements of array and also display the array elements in reverse order. 2. Write a program to sort a given array in ascending order and find the largest and second largest elements 3. Write a program to read two matrixes then multiply them after checking the condition for matrix multiplication.		
CO4	Develop C programs using Pointers, String handling functions and user-defined functions.		
M4.01	Make use of Pointers in programming	3	Applying
M4.02	Implement string manipulation functions - strcpy(), strcat(), strlen(), strcmp().	3	Applying
M4.03	Solve problems using 'User-defined functions'	3	Applying
	Suggested Experiments 1. Write a program to find biggest among three numbers using pointer. 2. Write a program to concatenate two given strings using built in functions 3. Write a program to find the diagonal sum, sum of columns, sum of arrays and sum of elements of a matrix using user defined functions.		
	Series Test 2	1.5	

**** - Suggested Open Ended Projects**

(Students have to do open ended experiments during the course for the purpose of continuous evaluation. This experiment shall be included in the bona-fide record.)

Example: Develop program such as

- Student CGPA calculator,
- Consumer billing,
- Candidate list sorting

Text / Reference:

T/R	Book Title/Author
T1	Let Us C – Yashavant Kanetkar, BPB Publications
R2	Programming in ANSI C, E. Balagurusamy, Tata McGraw Hill
R3	A Text Book on C, E. Karthikeyan , PHI Learning Pvt. Ltd
R4	Programming in C D. Ravichandran, New Age International Publishers
R5	C. Programming Language, Brian W. Kernighan and Dennis M. Ritchie, Prentice Hall

Online Resources:

Sl.No	Website Link
1	http://freecomputerbooks.com/The-C-Programming-Language.html
2	https://www.tutorialspoint.com/cprogramming/index.htm
3	https://www.guru99.com/c-programming-tutorial.html
4	https://www.cprogramming.com/
5	http://www.zentut.com/c-tutorial/

Sample Questions to Test Outcomes

1. Implement a C program to find sum and average of all elements in an integer array of size MxN where M,N and array elements can be input while the program is executing.
2. Implement a program to check whether the given string is a Palindrome or not.
3. Write a program to accept a string and count the number of vowels present in this string